

Final Exam: Micro 629 - Fall 2010

Texas A&M University

December 13th, 2010

By writing down your name you pledge to obey TAMU Honor Code:

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

Name (in pen, not pencil): _____

EXAM RULES

- Your interpretation of the questions makes part of the exam. Thus, no questions will be answered during the exam.
- No personal items (bags, cellphone, books, notebooks, calculators, etc.) will be allowed with you during the exam.
- Some of the numerals in the different problems require to prove the preceding numerals in the problem; if you cannot prove one, move on, and assume that it is true; use it to prove what is next and then come back if you have time.
- You can use results from class or homework whenever: (i) you make explicit reference to them and precisely write the result you are using, (ii) it is not the exact problem I am asking you to solve (in such a case I will ask you to “explicitly prove the statement”).

1. (25 points) Consider a production set $Y \subseteq \mathbb{R}^L$. Let π^Y be the profit function associated with set Y , i.e., for each $p \in \mathbb{R}_+^L$,

$$\pi^Y(p) \equiv \max_{y \in Y} p \cdot y.$$

- (a) (5/25) State the definition of free disposal and no free lunch. Draw a production set that satisfies these two properties. Illustrate the profit maximization for some price vector p .
- (b) (10/25) Is π^Y concave, convex, or neither of them? Provide an explicit argument.
- (c) (10/25) Suppose now that a firm with production set Y may face an uncertain market. The generic distribution of prices in the market is F , with density f . The firm is an expected profit maximizer, i.e., maximizes the function $\mu_{\pi^Y} \equiv \int \pi^Y(p)f(p)dp$. There are two markets in which the firm may participate.
 - Market 1: price is uncertain with some probability distribution.
 - Market 2: price is certain; it is exactly the expected price in Market 1.

Argue that the firm always finds participating in Market 1 as good as participating in Market 2.

2. (25 points) Let $Z \equiv \{z_1, \dots, z_n\}$ be a finite set of prizes and $\succsim$ a preference on $\Delta(Z)$, i.e., the set of lotteries on Z . The generic lottery is denoted $\pi \in \Delta(Z)$.

- (a) (5/25) State the definition of the Independence axiom (IA), completeness, and transitivity.
- (b) (10/25) Assume that $\succsim$ is complete, transitive, and satisfies IA. Prove that for each $\alpha \in [0, 1]$, if $\pi \succsim \pi'$, then $\pi \succsim \alpha\pi + (1 - \alpha)\pi' \succsim \pi'$.
- (c) (10/25) Prove that $\succsim$ is complete, transitive, continuous, and satisfies IA if and only if there is a function $v : Z \rightarrow \mathbb{R}_{++}$ such that $\succsim$ is represented by the function that associates with $\pi \in \Delta(Z)$ the value

$$\Psi(\pi) \equiv v(z_1)^{\pi(z_1)} \times \dots \times v(z_n)^{\pi(z_n)}.$$

3. (25 points) Let $e \equiv ((\succsim^i)_{i \in N}, (\omega^i)_{i \in N}) \in \mathcal{E}_p$ be an exchange economy.

- (a) (5/25) State the definition of feasible allocation for e
- (b) (5/25) State the definition of competitive equilibrium for e .
- (c) (5/25) State the definition of Pareto efficient allocation for e .
- (d) (10/25) Explicitly prove that if preferences are locally non-satiated, then each competitive equilibrium outcome for e is Pareto efficient.

4. (25 points) Let $\succsim$ be a complete, transitive, and continuous preference on $\mathbb{R}_+^L$. Let $x \in \mathbb{R}_+^L$, $p \in \mathbb{R}_+^L$, and $w \equiv p \cdot x$. Suppose that $w > 0$ and that for each $y \in U(\succ, x)$, $p \cdot y \geq w$. Prove that x maximizes $\succsim$ in $B(p, w)$, i.e., $x \in C^{\succsim}(p, w)$.

5. (Bonus 25 points) Suppose that $N \equiv \{1, \dots, n\}$. Let $e \equiv ((\succsim^i)_{i \in N}, \Omega) \in \mathcal{E}_c$ be such that $\Omega \gg 0$ and for each pair of agents $\{i, j\} \subseteq N$, $\succsim^i = \succsim^j$. Prove that if preferences are convex and strictly monotone, then $(\frac{1}{n}\Omega, \dots, \frac{1}{n}\Omega)$ is a Pareto efficient allocation for e .